Architecture diagram

Rendered PNGs: `arch_component.png` (component/coordination view) and `arch_flow.png` (probe/antenna information-flow view) are in `docs/`. The PDF is built from those images — see the pandoc command at the bottom.

Component / coordination view

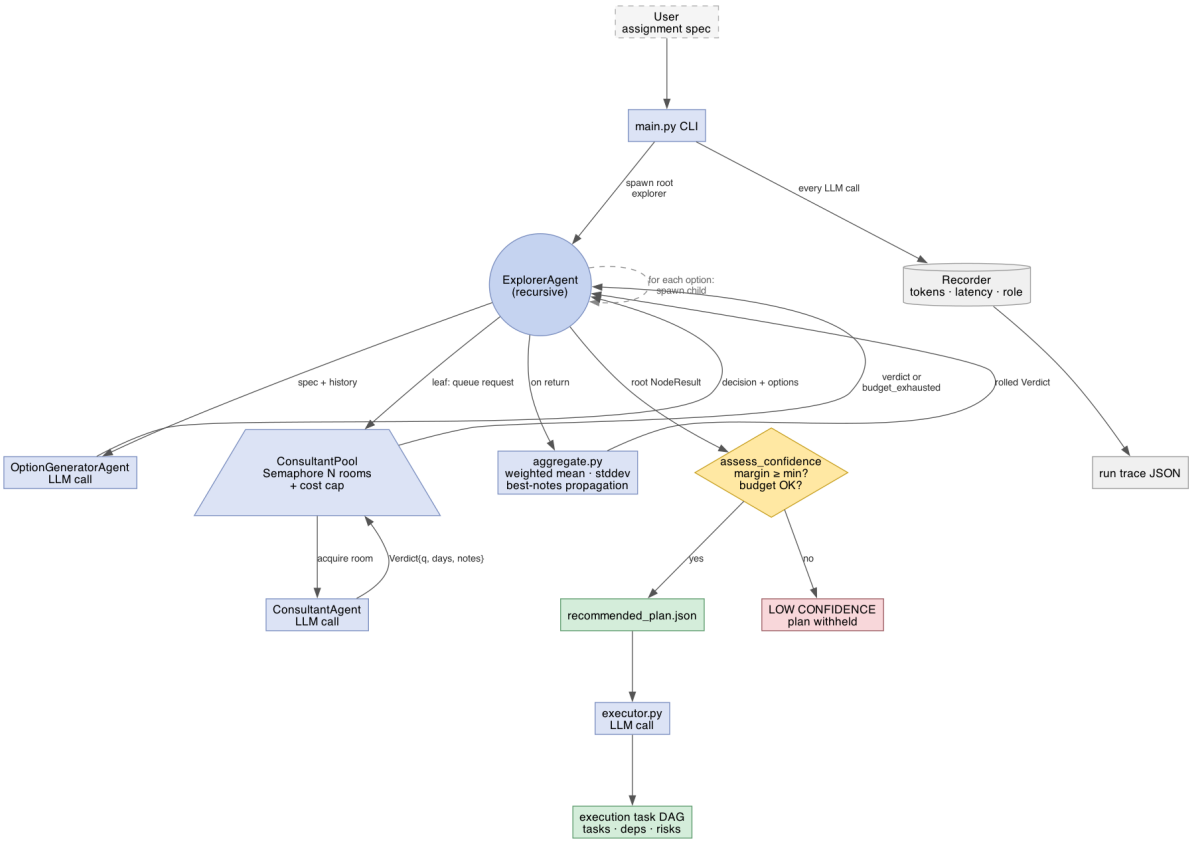

Figure 1: Component and coordination view

```
flowchart TD
    User[User: assignment spec] --> Main[main.py CLI]
    Main -->|spawn root explorer| Explorer((ExplorerAgent<br/>recursive))
    Explorer -->|"(spec, history)"| OptGen[OptionGeneratorAgent<br/>LLM call]
    OptGen -->|"decision + options[]"| Explorer

    Explorer -->|"if leaf:<br/>request(history)"| Pool[/ConsultantPool<br/>Semaphore N rooms<br
    Pool -->|acquire room| Consult[ConsultantAgent<br/>LLM call]
    Consult -->|"Verdict{q, days, notes}"| Pool
    Pool -->|verdict<br/>or budget_exhausted| Explorer

    Explorer -->|"for each option:<br/>spawn child explorer"| Explorer
```

```
    Explorer -->|on return| Agg[aggregate.py<br/>weighted mean + stddev<br/>best-notes propaga
    Agg -->|rolled Verdict| Explorer

    Explorer -->|root NodeResult| Gov{assess_confidence<br/>margin >= min?<br/>budget OK?}
    Gov -->|yes| Plan[recommended_plan.json]
    Gov -->|no| Refuse[LOW CONFIDENCE<br/>plan withheld]

    Plan --> Exec[executor.py<br/>LLM call]
    Exec --> Tasks[execution task DAG<br/>tasks + deps + risks]

    Main -->|every LLM call| Rec[(Recorder<br/>tokens, latency,<br/>per-role)]
    Rec --> Trace[run trace JSON]
    Explorer --> Trace
```

Information-flow view (probe / antenna mental model)

```
flowchart TB
    Root[Root antenna<br/>history=[]] -->|spawn probe| O1[Option A child]
    Root -->|spawn probe| O2[Option B child]
    Root -->|spawn probe| O3[Option C child]

    O1 -->|spawn| O1a[A→a leaf]
    O1 -->|spawn| O1b[A→b leaf]
    O2 -->|spawn| O2a[B→a leaf]
    O2 -->|spawn| O2b[B→b leaf]
    O3 -->|spawn| O3a[C→a leaf]
    O3 -->|spawn| O3b[C→b leaf]

    O1a -->|consultant verdict| Pool[(ConsultantPool)]
    O1b --> Pool
    O2a --> Pool
    O2b --> Pool
    O3a --> Pool
    O3b --> Pool

    Pool -->|verdicts| O1a
    Pool --> O1b
    Pool --> O2a
    Pool --> O2b
    Pool --> O3a
    Pool --> O3b

    O1a -->|verdict up| O1
    O1b --> O1
    O2a --> O2
    O2b --> O2
    O3a --> O3
```

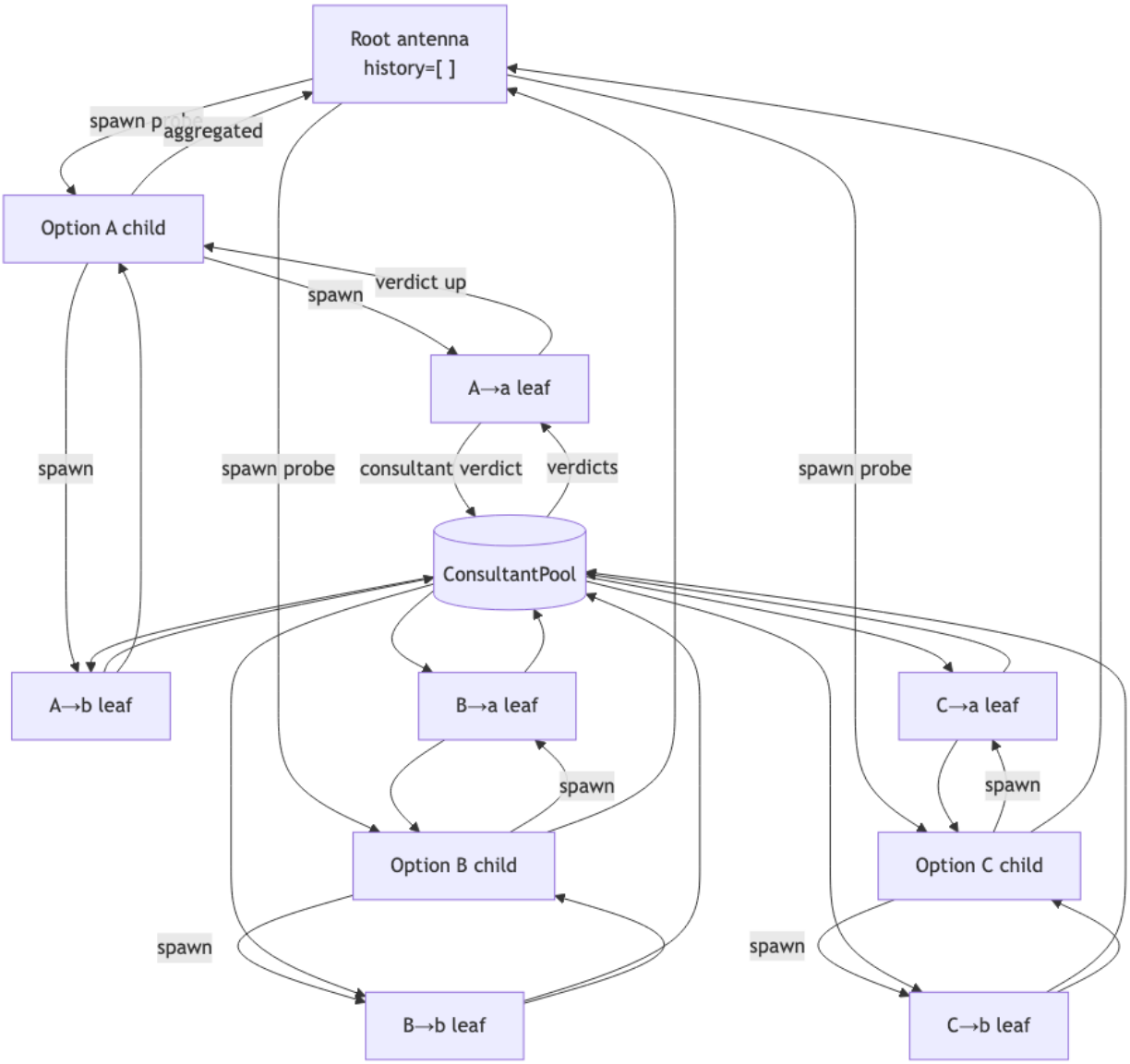

Figure 2: Information-flow view: probe / antenna mental model

```
O3b --> O3

O1 -->|aggregated| Root
O2 --> Root
O3 --> Root
```

Coordination notes

- **Fan-out** is implemented with `asyncio.gather`; thousands of in-flight explorer coroutines are cheap.
- **Bottleneck and backpressure** live in the consultant pool. The semaphore caps in-flight LLM calls; `max_consultations` caps total cost. Both the explorer tree and any future executor wait at this single chokepoint.
- **Aggregation** happens at every antenna on its way back up. The rolled `Verdict` is consultation-weighted so deeper subtrees get proportional influence, and the best leaf's feasibility notes are preserved up the chain so reviewers see real consultant text at the root.
- **Governance** is a pure function (`assess_confidence`) over the root's branches plus the propagated `budget_exhausted` flag — it never mutates the tree, only decides whether to publish a recommendation downstream.

Human-in-the-loop points

- The user picks `--rooms`, `--depth`, `--options`, `--max-consultations`, `--min-margin` per run.
- The user reviews the per-option breakdown printed at the end and the JSON trace before passing the recommended plan to the executor.
- The executor's task DAG is intended for human review before any downstream agent is allowed to execute it.